Reaction Sheet

CBSE Class 12 Chemistry • NCERT Part II, Chapters 6–10

Blue boxes: NCERT syllabus. Purple boxes: JEE extras.

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Chapter 6 Haloalkanes and Haloarenes

Preparation

$R-OH + HX \xrightarrow{ZnCl_2} R-X$; $R-OH + PCl_5 \longrightarrow RCl + POCl_3 + HCl$; $3\,ROH + PCl_3 \longrightarrow 3\,RCl + H_3PO_3$; $ROH + SOCl_2 \longrightarrow RCl + SO_2 + HCl$

Alkene + HX (Markovnikov); + HBr/peroxide (anti-Markovnikov). Free-radical halogenation ($h\nu$).

Finkelstein: $RCl + NaI \xrightarrow{\text{dry acetone}} RI + NaCl$. **Swarts:** $CH_3Br + AgF \longrightarrow CH_3F + AgBr$.

Sandmeyer: $ArN_2{}^+X^- \xrightarrow{Cu_2Cl_2/HCl} ArCl$; $\xrightarrow{Cu_2Br_2/HBr} ArBr$. Arenes: $ArH + Cl_2 \xrightarrow{FeCl_3} ArCl$.

Reactions

aq. KOH	ROH (substitution)
alc. KOH	alkene (elimination, Saytzeff)
KCN / AgCN	RCN / RNC
KNO_2 / $AgNO_2$	R–ONO / R–NO_2
NaOR′	ROR′ (Williamson)
Mg, dry ether	RMgX (Grignard; with H_2O gives RH)
Wurtz: 2RX + 2Na (dry ether)	R–R
Wurtz–Fittig: ArX + RX + Na	Ar–R
Fittig: 2ArX + 2Na	Ar–Ar
Chlorobenzene + NaOH, 623 K, 300 atm	phenol (Dow)

Chlorobenzene + $Cl_2/FeCl_3$, o- and p-products
HNO_3/H_2SO_4, $CH_3Cl/AlCl_3$
S_N2: $CH_3 > 1° > 2° > 3°$; S_N1: $3° > 2° > 1° > CH_3$ (benzylic, allylic fast). Leaving group: I > Br > Cl > F.

JEE extra

Benzyne ($NaNH_2$); DDT, freons, $CHCl_3 \rightarrow COCl_2$ (phosgene) in light; iodoform from ethanol/acetone with $I_2/NaOH$; E2 with bulky base gives Hofmann product.

Chapter 7 Alcohols, Phenols and Ethers

Preparation

Alcohols: alkene + H_2O/H^+ (Markovnikov); hydroboration–oxidation (anti-Markovnikov); RCHO/RCOR + $NaBH_4/LiAlH_4$ or H_2/Pd; RCOOH + $LiAlH_4$; Grignard: HCHO → 1°, RCHO → 2°, ketone → 3°.

Phenol: benzenesulphonic acid + NaOH (fusion) then H^+; chlorobenzene + NaOH (623 K, 300 atm); diazonium + warm H_2O; cumene + O_2 → hydroperoxide → phenol + acetone.

Ethers: $2\,C_2H_5OH \xrightarrow{H_2SO_4,\,413\,K} C_2H_5OC_2H_5$; Williamson: RX + R′ONa → ROR′ + NaX (RX primary).

Reactions

ROH + Na	RONa + H_2 (acidic)
Esterification	$ROH + R'COOH \overset{H^+}{\rightleftharpoons} R'COOR + H_2O$
Dehydration (conc. H_2SO_4, 443 K)	alkene ($3° > 2° > 1°$)
Oxidation	1° → aldehyde (PCC) → acid ($KMnO_4$); 2° → ketone (CrO_3); Cu/573 K: 3° → alkene
Phenol + Br_2(aq)	2,4,6-tribromophenol (white ppt); Br_2/CS_2: o- and p-bromophenol
Phenol + dil. / conc. HNO_3	o-/p-nitrophenol / picric acid
Kolbe	PhONa + CO_2 (400 K, 4–7 atm), H^+ → salicylic acid
Reimer–Tiemann	PhOH + $CHCl_3$ + NaOH, H^+ → salicylaldehyde
Phenol + Zn dust	benzene
Phenol + $Na_2Cr_2O_7/H_2SO_4$	benzoquinone
Ether + HI	ROH + R′I (smaller/1° R′ gets I; anisole → phenol + CH_3I)
Anisole	o/p substitution: bromination, nitration, Friedel–Crafts

Tests

Test	Positive for	Observation
Lucas (conc. HCl + $ZnCl_2$)	3°/2°/1° alcohol	turbidity at once / 5 min / only on heating
Neutral $FeCl_3$	phenol	violet colour
Bromine water	phenol	white ppt
Iodoform (I_2/NaOH)	$CH_3CH(OH)-$ (ethanol, propan-2-ol)	yellow CHI_3

Chapter 8 Aldehydes, Ketones and Carboxylic Acids

Preparation

1° ROH + PCC → RCHO; 2° ROH + CrO_3 → ketone. **Rosenmund:** $RCOCl + H_2 \xrightarrow{Pd-BaSO_4}$ RCHO. **Stephen:** RCN + $SnCl_2$/HCl, H_3O^+ → RCHO. Nitrile + DIBAL-H → aldehyde. **Etard:** toluene + CrO_2Cl_2 → benzaldehyde. **Gatterman–Koch:** benzene + CO/HCl ($AlCl_3$) → benzaldehyde. Friedel–Crafts acylation → aryl ketone. $RCOCl + R'_2Cd$ → ketone. RCN + R′MgX → ketone.

Acids: 1° ROH/RCHO + $KMnO_4$; alkylbenzene + $KMnO_4$/KOH → benzoic acid; RCN + H_3O^+; RMgX + CO_2 then H_3O^+; ester/amide hydrolysis.

Reactions of aldehydes and ketones

HCN	cyanohydrin NaHSO$_3$: bisulphite adduct
ROH (dry HCl)	hemiacetal → acetal; ketone + ethylene glycol → ketal
NH_2–Z	NH_2OH: oxime; NH_2NH_2: hydrazone; 2,4-DNP: orange ppt; semicarbazide: semicarbazone
Reduction	$NaBH_4$/$LiAlH_4$: alcohol; **Clemmensen** (Zn–Hg/HCl), **Wolff–Kishner** (NH_2NH_2/KOH, glycol): CH_2
Aldol (dil. NaOH, needs α-H)	β-hydroxy carbonyl → α,β-unsaturated carbonyl
Cannizzaro (conc. NaOH, no α-H)	alcohol + carboxylate salt
Benzaldehyde	m-substitution (nitration)

Reactions of carboxylic acids

$RCOOH + NaHCO_3 \longrightarrow RCOONa + CO_2 + H_2O$; $PCl_5/SOCl_2$ → RCOCl; NH_3, heat → amide; $LiAlH_4$ → RCH_2OH; P_2O_5 → anhydride.
Decarboxylation: RCOONa + NaOH/CaO → RH. **Kolbe electrolysis:** 2RCOONa → R–R.
HVZ: $RCH_2COOH \xrightarrow{X_2/\text{red P}}$ RCHXCOOH. Benzoic acid: m-substitution.

Tests

Test	Positive for	Observation
Tollens'	all aldehydes	silver mirror
Fehling's	aliphatic aldehydes	red ppt of Cu_2O
Iodoform	CH_3CO- (methyl ketones, ethanal)	yellow ppt
2,4-DNP	aldehydes and ketones	orange/yellow ppt
$NaHCO_3$	carboxylic acids	CO_2 effervescence
Ester test	acid + alcohol, conc. H_2SO_4	fruity smell

Chapter 9 Amines

Preparation

Nitro reduction: $ArNO_2 \xrightarrow{Fe/HCl \text{ or } H_2/Pd} ArNH_2$. Ammonolysis: $RX + NH_3 \rightarrow$ mixture of 1°, 2°, 3°, quaternary salt.
Nitrile / amide reduction: $RCN \xrightarrow{LiAlH_4} RCH_2NH_2$ (one C more); $RCONH_2 \xrightarrow{LiAlH_4} RCH_2NH_2$.
Gabriel phthalimide: pure 1° aliphatic amines. **Hoffmann bromamide:** $RCONH_2 + Br_2 + 4\,NaOH \longrightarrow RNH_2$ (one C less).

Reactions

Acylation ($(CH_3CO)_2O$ / RCOCl)	amides (acetanilide); 3° no reaction
Carbylamine ($CHCl_3$ + alc. KOH)	1° amines → isocyanide (foul smell)
HNO_2	1° aliphatic: alcohol + N_2; aromatic (273 K): diazonium salt
Hinsberg ($C_6H_5SO_2Cl$)	1°: soluble in alkali; 2°: insoluble; 3°: no reaction
Aniline + Br_2(aq)	2,4,6-tribromoaniline (white ppt)
Aniline + conc. H_2SO_4, 453–473 K	sulphanilic acid
Nitration	o-, p- and m-nitroaniline (protect by acetylation for p)

Diazonium salt reactions

Cu_2Cl_2/HCl	ArCl (Sandmeyer)	Cu/HCl	ArCl (Gattermann)
CuCN/KCN	ArCN	KI	ArI
HBF_4, heat	ArF	H_2O, warm	ArOH
H_3PO_2 or C_2H_5OH	ArH	phenol (OH^-)	orange azo dye
aniline (H^+)	yellow p-aminoazobenzene		

Basicity (water): $(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2 > NH_3 > C_6H_5NH_2$; $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$.

Chapter 10 Biomolecules

Key facts

Glucose	$C_6H_{12}O_6$, aldohexose, pyranose ring, α/β anomers at C1
Fructose	ketohexose, furanose ring
Sucrose	glucose + fructose (C1–C2 link), non-reducing; hydrolysis gives invert sugar
Maltose / lactose	glucose + glucose (α 1,4) / galactose + glucose (β 1,4); reducing
Starch / cellulose / glycogen	amylose + amylopectin (α); β-glucose chains; highly branched animal starch
Proteins	α-amino acids joined by peptide bonds (–CO–NH–); 1°, 2° (α-helix, β-sheet), 3°, 4° structure
Vitamins	fat-soluble A, D, E, K; water-soluble B, C. A: night blindness; B_1: beri-beri; B_{12}: pernicious anaemia; C: scurvy; D: rickets; E: muscle weakness; K: delayed clotting
Nucleic acids	nucleotide = base + pentose + phosphate; DNA: deoxyribose, A–T, G–C, double helix; RNA: ribose, U for T; mRNA, tRNA, rRNA

JEE extra

Glucose with phenylhydrazine gives osazone (glucose, fructose, mannose give the same osazone). Isoelectric point of amino acids. Enzymes: specific biocatalysts, lower E_a.